

THE NAUTILUS.

VOL. XXXI. OCTOBER, 1917. NO. 2

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**THE
NAUTILUS
A QUARTERLY
DEVOTED TO THE INTERESTS
OF CONCHOLOGISTS**

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OCTOBER, 1917.

No. 2

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NOTES ON THE VARIATION OF ISCHNOCHITON CONSPICUUS CPR.

BY E. P. CHACE.

To the naturalist the study of variation and environment and their relation to each other is always interesting and to the conchologist who studies his shells in their natural surroundings as well as in the cabinet many things are revealed. It is not, often, however, that variation in form may be so easily traced to qualities of environment as in the following instance.

Ischnochiton conspicuus, Cpr. is common at San Pedro and is usually found on the under side of rocks in sandy tidepools. In this situation they grow quite regularly, showing but slight variation in form. They are active fellows and evidently sensitive to light; for if the rocks to which they are clinging are turned over, they soon glide away, always taking the shortest route to the under side of the rock.

While cleaning a lot of this species taken at San Pedro last fall two specimens were noticed which differed so widely from the others that they might easily have been mistaken for another species. They were much wider and lower-arched than the typical form and the posterior corners of the valves were rounded off, making the lateral areas very narrow.

Hoping to find more specimens of this odd form, a trip to Point Firmen was devoted wholly to the collection of chitons, with interesting results. The usual species were found in the tidepools including numerous specimens of *Ischnochiton conspicuus* of the ordinary form. Ledges of soft rock beyond the tidepools contained many old pholad holes and in these were found the form for which I was searching. Twenty specimens of various sizes were collected from as many pholad holes. In the larger specimens the foot had become so greatly enlarged to fit the concave

bottoms of the holes that it was impossible for them to curl up in the usual manner. Some of these specimens were so badly eroded by the sand and gravel which wash in and out of the holes that the anterior valve was reduced to two thirds of its normal height.

In color pattern, sculpture, and mantle characters these specimens were identical with those from the tidepools, and, as will be seen by referring to the table of measurements, the smaller specimens approach quite closely to the proportions of the typical or tidepool forms.

TYPICAL OR TIDEPOOL FORMS. (Lot 1)				SPECIMENS FROM PHOLAD HOLES. (Lot 2)			
<i>Lon.</i>	<i>Lat.</i>	<i>Index.</i>	<i>Div.</i>	<i>Lon.</i>	<i>Lat.</i>	<i>Index.</i>	<i>Div.</i>
99	35	2.52	130°	65	33	1.97	135°
83	33	2.51	130°	63	32	1.97	155°
83	30	2.76	125°	62	27	2.29	140°
62	22	2.80	125°	60	30	2.00	130°
59	23	2.56	130°	56	29	1.93	130°
38	14	2.71	130°	42	17	2.47	130°

In brief, Lot 1 shows an angle of divergence constant at 125° to 130°, where Lot 2 shows an angle varying from 130° to 155°, and a proportion of length to width 2.51 to 2.80 as against a proportion ranging from 2.47 to 1.93.

The noticeable differences to the eye are first, the narrow and sharply raised lateral areas, and second, the shape of the posterior edge of the median valves. In the tidepool specimens the posterior or exposed edge of each valve is a straight line, while in specimens from the pholad holes this line becomes a double convex curve, the most posterior portion of the valves being about midway between the beaks and the girdle.

These differences seem to be explained by the following facts. In collecting, the tidepool specimens are usually found on the under side of large rocks and well back from the edge. This situs protects them from the light which they evidently find objectionable, but it makes necessary a nightly journey of about two feet to the nearest growth of algae on which they feed. This activity stretches the girdle downward from the edges of the valves and permits a free play of all the valves so that the mantle deposits its shelly secretions according to the normal habit of the species. The specimens living in the pholad holes, however, apparently never leave them as they are frequently found feeding on the fucus which overhangs them. It

protects them from the light, so they have no occasion to move about, and the sand which is washed down into these burrows would make re-entrance almost impossible. A series of these specimens shows a gradual change of form. The young specimens are very similar to young specimens from the tidepools, but as they increase in size they become crowded so that the valves press against each other, especially at the posterior end where the valves are bent back across the bottom of the hole. This crowding of the valves upon each other and the crowding of the girdle against the outer edges of the valves so displaces portions of the mantle as to cause the changes noted above.

Several specimens from each situs were disjointed and a study of the individual valves showed that those from pholad-hole specimens were thicker and had shorter sutural plates and a wider sinus, this last being especially noticeable in the valves from the posterior end. Apparently this change in the sinus is the result of the broadening of the connecting ligaments due to compression by the crowding valves.

A count of the insertion plates of these disjointed specimens was made and considerable variation noticed. So much, in fact, that more specimens were pulled apart for the express purpose of counting these plates. Representative counts were as follows: 9 slits on the anterior valve, 2-3 on the median valves, and 10 on the posterior. Others show 12, 2-3, 8; 11, 2-3; 14, 3-4, 11. Absolutely no difference in this character could be found between specimens from the tidepools and those from the pholad holes.

On page 64 of vol. xiv of the Manual of Conchology, Dr. Pilsbry says, "Carpenter has given a varietal name to a broad, worn specimen which he thus describes:

"Var. *solidus*. Very solid, wide, ashen; inside whitish, the posterior valve with 10, central valve 2-3, anterior valve 12 slits. Length 72, breadth 40, divergence 130°. Carpenteria, near Sta. Barbara, Cal. This is scarcely more than an individual variation. The mantle (girdle) is normal. The sculpture is worn away except at the edge. It has evidently lived in a very exposed situation."

From this description and the figure which he gives of the valves it would seem that this is the same form as my specimens from the pholad holes. I fully agree with Dr. Pilsbry that it is hardly worthy of varietal rank. It is, however, too distinct in appearance to be labeled simply *Ischnochiton*

conspicuus, Cpr., and I have therefore marked these specimens from the pholad holes *I. conspicuus*, Cpr., *form solida*, Cpr., using the term *form* as advocated by Dr. Cockerell, “to designate variations plainly due to environment.”^[1]

LAMPSILIS VENTRICOSA COHONGORONTA IN THE POTOMAC RIVER.

BY WILLIAM A. MARSHALL.

In 1912 Dr. A. E. Ortmann recorded^[2] finding in the Potomac River a variety of *Lampsilis ventricosa*^[3] to which he gave the name *cohongoronta*. His records were:

September 4, 1909. Potomac River, Hancock, Washington, Co., Md. (about two dozen).

May 9, 1911. South Branch, Potomac River, Southbranch, Hampshire Co., W. Va. (about a dozen).

August 16, 1911. Shenandoah River, Harper's Ferry, Jefferson Co., W. Va. (a single male, below medium size).

May 6, 1912. South Branch, Potomac River, Romney, Hampshire Co., W. Va. (about a dozen).

Dr. Ortmann remarked "It is probable that this species will turn up elsewhere in the Potomac. The localities known at present are all to the west of the Blue Ridge Mountain, that is to say, within the Great Alleghany Valley and the Alleghany Mountains."

Perhaps the above prediction has been realized in a specimen found at Great Falls, Md., by Mr. Manly D. Barber of Knoxville, Tennessee, in Sept. 1915. At that time Mr. Barber brought to the National Museum a basketful of naiades which he had collected the same day at Great Falls, about 18 miles above Washington. Among the shells, which were mostly dead ones, was a specimen of *cohongoronta*, dead, but in a fine state of preservation and with the periostracum nearly unblemished except for the usual erosion at the beaks. Its appearance indicated that it had been recently alive and that

its home had been in the immediate vicinity of the place in which it was found. Had it been washed down from Harper's Ferry, some 50 or more miles above Great Falls it probably would have shown ill effects from so long a journey.

When found the two valves were separated, but so accurately do they fit together that it is evident they belong to the same individual. The fact that the valves were separated and yet were found near each other is additional (though not conclusive) evidence that they had not been transported any great distance by currents. At any rate this is the first recorded finding of the species in the Potomac River so far south as Great Falls.

The specimen is rather a small one. It measures, length 71 mm.; height 47 mm.; diameter 28 mm. It is in the collection of the U. S. National Museum, catalogue number 273834.

COLLECTING DAYS ABOUT THE NAVAL STATION, GUANTANAMO BAY, CUBA.

BY JOHN B. HENDERSON.

In March last, while waiting for a boat to take us to Haiti, Dr. Bartsch and I spent nearly three weeks at the U. S. Naval Station at the entrance to Guantanamo Bay. We employed our time in exploring the country about and subjecting it to a high degree of intensive collecting. In this eastern corner of Cuba the coastal strip of some ten miles in width is a semi-arid region with a complex of mountains that are either quite bare of trees or, at most, covered with a scrub forest and low-growing spiny shrubs, with, here and there, a wealth of cacti that almost suggests Lower California. The rock foundation of all this region,—barring some shore strips of very recently elevated coral, is everywhere composed of about everything in the line of rocks except limestone. This is a condition that in the Antilles usually spells disappointment and failure to the snail hunter. North of the big bay and then across several miles of low flat country, just where the foothills of the sierras begin, lies the city of Guantanamo, interesting to us as the home of Charles Ramsden, the naturalist. Just north of Guantanamo is a great rampart of high limestone mountains which beckon most alluringly to the collector. Sections of this rampart, somewhat arbitrarily marked off, are the “Monte Verde,” the “Monte Toro” and the “Monte Libano” of classic fame in Cuban Natural History.

In company with Ramsden we spent a wonderful day on nearby Monte Libano but a revolution that was then devastating the province and filling the land with incendiaries and bandits drove us out of this richer field and obliged us to confine our attentions thereafter to the arid country lying within the safer limits of the Naval Station,—some fifty square miles upon which Uncle Sam holds a long lease.

It seems to be a natural law that arid or desert lands support but few species of snails, but that these few species exist in great numbers and that they take on a very considerable range of variation. All this is perfectly true of this region. We were constantly amazed by the great number of specimens to be found; and each day of exploration in some new valley or over some range of hills added even greater figures of abundance to our already astonishing records.

The “prevailing” snail of this region is *Cepolis ovumreguli* Lea. Its shell is very suggestive of the true helix of Spain or Algeria of the *lactea* group. The variation is exceedingly great in color, size and shape, and it would make a dozen excellent species if the intermediates were left out of account. Those living near the coast and among the cacti of the most arid parts of the district are of whiter and more dull color, are more banded and show a decided tendency to abnormalities, especially about the apertures. Specimens from further inland are more polished and shining, even as though varnished, and are much more given to a dotted or fly-specked type of ornamentation than to bands. A fence-post or a dead tree-limb with a hundred specimens closely assembled in aestivation was no unusual sight. We learned finally to pay no attention to them. Upon the low bushes in certain localities the lovely little *Cepolis lucipeta* Poey cling like berries. These are the largest and finest of the species I have ever seen. The range of color variation in this delightful little snail is also very great, but the colors never blaze out in the vivid flash of the *Polymitas*. The blues and purples and chestnut browns are subdued but very rich and splendid. One very noticeable color form is the subsp. *velasqueziana* of Poey where the many broken bands of the type coalesce into two broad bluish-black zones of solid color.

As nearly all the vegetation of this dry region bears thorns we did not at first discover that many of these thorns were in reality *Macrocerami*. When we did find this out we could see nothing else. Bartsch and I finally agreed, and shook hands upon it, that we would gather no more of them, and a stiff penalty was placed upon any violation of the compact. Two hundred and more from one bush is an earlier record before we really got started. This is the *Macroceramus festus* (Gundl.) Pfr., blue and yellow and buff in color. Another arboreal snail of this section is *Polymita versicolor* Born and it is probably very abundant in places although we never saw more than fifteen or twenty on any one tree. This is to me the least attractive species of that

wonderful genus of richly painted snails. The brilliant yellow and pink are too primitive and the two colors do not seem to harmonize very well. It always impresses me as an experimental species that was laid aside in nature's laboratory as not wholly a success.

There were some ground snails too, but to secure living ones required much grubbing up of tufts of tall grass and shaking out their roots, like digging up miniature potatoes. These are the *Annularia putris* (Gundl.) Pfr. and the *Chondropoma marginalbum* (Gundl.) Pfr., the latter apparently quite rare. There are no minute things beyond some few *Thysanophora inaguensis* Weinland.

Some days we spent gathering marines on the little pebbly beaches hidden far down under the lofty cliffs that mark this rugged shore line, and we obtained some unusual species washed up from the exceedingly narrow island-shelf; blue water is but a few hundred yards out. Among these are some *Conus cedonulli* Lam. Beach collecting is, however, an aggravation; and too much of it becomes a misdemeanor in the collector's ethical code, for it obliges an acceptance of something short of the best.

AMNICOLIDÆ FROM ONEIDA LAKE, N. Y.

BY HENRY A. PILSBRY.

The New York College of Forestry, under the direction of Professor Hugh P. Baker, is carrying on a biological survey of Oneida Lake and has issued an interesting bulletin^[4] upon the relations of mollusks to fish, by Frank C. Baker. Some *Amnicolidæ* obtained during this work, and subsequent to the preparation of the bulletin were submitted to the writer. The collection proves to be of considerable interest, including some species not before noticed.

AMNICOLA BAKERIANA, n. sp.

The shell is umbilicate, turritid-conic, thin, whitish-corneous, somewhat translucent, with unevenly developed striation, distinct, and close in places, weaker and sparse elsewhere. The summit is decidedly obtuse, as in *A. limosa*, the first whorl being nearly planorboid; subsequent whorls are evenly, strongly convex. The aperture is very shortly ovate, almost round, its length contained almost $2\frac{1}{2}$ times in that of the shell. Peristome thin, in contrast with the preceding whorl for a short distance.

Length 4.3, diam. 2.7 mm.; 5 whorls (type).

Length 3.75, diam. 2.3, length of aperture 1.35 mm.; $4\frac{2}{3}$ whorls.

Length 4.1, diam. 2.75, length of aperture 1.65 mm.; $4\frac{2}{3}$ whorls.

Oneida Lake; off Short Point in $8\frac{1}{2}$ ft., mud bottom. Lower South Bay, in 18 ft., on mud bottom.

This species resembles *A. limosa* in the conspicuously obtuse apex, but differs by the more elevated, turritid spire and the smaller calibre of the whorls, hence smaller aperture. It is also a weaker shell, with more whorls in specimens of the same length.

There is also an abundant smaller form, resembling the typical form in texture, apex and shape of the whorls, varying in form, but relatively broader than the type. There are some intermediate examples, but as Mr. Baker considers it desirable to have a designation for this form, it may be called *A. bakeriana* form *nimia*. The type measures: length 3, diam. 2.5, length of aperture 1.4 mm.; 4 whorls.

AMNICOLA CLARKEI, n. sp.

The shell is narrowly umbilicate, conic, a little obtuse at the apex, corneous, nearly smooth. The whorls are very convex, separated by a deep suture, the last whorl tubular. The aperture is distinctly oblique, almost circular, the upper end rounded, but a trifle more narrowly so than the base. It projects but little beyond the preceding whorl laterally. The peristome is thin, continuous, scarcely or barely in contact with the preceding whorl above.

Length 3.1, diam. 1.9, length aperture 1.1 mm.; 5 whorls (type).

Length 2.8, diam. 1.6, length aperture 0.85 mm.

Operculum having the spiral rather large, the nucleus being above the lower third.

This little species resembles *Lyogyrus* by its tubular whorls of small calibre. The whorls are more convex and increase less rapidly than in *Amnicola walkeriana*, which is also less slender. *A. schrockingeri* Ffld. has less deeply convex whorls, and the apex is more acute. *A. bakeriana* is much larger, with a more obtuse apex.

Found in Short Point Bay, Oneida Lake, near shore, in 3 feet of water, bottom of sand with algae; also in Lower South Bay, etc. Collected by Mr. F. C. Baker.

It is named for Dr. John M. Clarke, the distinguished Director of the Museum of the State of New York.

AMNICOLA ONEIDA, n. sp.

The shell is typically more slender than *A. lustrica*, turritid-conic, narrowly umbilicate, corneous, minutely striate. The apex is slightly obtuse, but the first whorl projects visibly, as in *lustrica*, whorls *very convex*, parted by a *deep suture*. The aperture is ovate, small, its length contained more than 3 times in that of the shell; upper extremity narrowly rounded. The peristome is continuous, thin, very briefly in contact with the preceding whorl above.

Length 4, diam. 2, length of aperture 1.25 mm.; 6 whorls.

Lower South Bay, Oneida Lake, N. Y., collected by F. C. Baker, 1916.

This species is typically narrower than *A. lustrica* Pils., with a smaller aperture and shorter whorls; but it is chiefly distinguished by the more convex whorls (deeper suture), and the rounded instead of angular posterior end of the aperture. In *Paludestrina nickliniana* the last whorl is much longer. Possibly it may be a subspecies of *lustrica*, yet it has so distinct an appearance that a special name seems desirable. There are also wider examples, which still differ from *lustrica* by the deeper suture and aperture.

NEW GENERA AND SPECIES OF CENTRAL AMERICAN NAIADES.

BY L. S. FRIERSON.

In 1893 Messrs Crosse and Fischer divided the Mexican Naiades into quite a number of sections, to which they assigned names. Almost simultaneously (in 1900) von Martens and C. T. Simpson, in treating the Central American Naiades, accepted some of these sections of Crosse and Fischer, raising them to generic or subgeneric rank. Because of paucity of material, considerable diversity of opinion concerning the specific identity of several species may be noted in the works of these authors. Furthermore, their work of classification being done independently and from different points of view, the same species was sometimes placed by them in different genera.

Thanks to the arduous labors of A. A. Hinkley, who has again and again enriched our cabinets with material and data from these tropical countries, we are enabled to offer the following suggestions concerning some of the genera of these shells, and also the description of an unpublished species.

NEPHRONAIAS. This genus has for its type the *Unio plicatulus*, Küster, a species identified by von Martens as belonging to the Lampsiline shells, as *aztecorum*. Mr. Simpson however believed it to be nearly allied to the *persulcatus*, a markedly Unioidean shell. In this the writer follows Mr. Simpson.

The genus *Nephronaias* as constituted by Mr. Simpson embraces two quite distinct groups, divisible as follows.

Nephronaias (s. s.) embraces *plicatulus*, *persulcatus*, *melleus*, *dysoni*, *ortmanni*, *ravistellus*, etc. Ample material of these two latter species show that they are anatomically very closely allied to *Elliptio*. There is no sexual

difference of shape, and the gill is gravid in its whole length. *Nephronaias* differs from *Elliptio* in its sulcated disc, in its beak sculpturing, etc.

Included in *Nephronaias* by Simpson are, however, shells of a totally different type, such as *medellinus*, *gundlachi*, *sapotalensis*, etc. These latter are sexually dimorphic, smoother, more generally rayed, and the gravid uterus is of Lampsiline type.

The position of the dorsal scars within the beak cavities is different, in the examples of the pseudo *Nephronaias* seen by the writer. *Nephronaias* (s. s.) possesses an (accessory?) adductor scar attached to the frontal portion of the cardinal teeth, which is either absent or obsoletely marked in the second assemblage. For this latter group the writer, therefore, proposes to use the generic term of *Actinonaias* Crosse and Fischer, 1893, type *U. sapotalensis* Lea. The female of this species has been described by Dr. Ortmann (1912). *Actinonaias* embraces, besides the type, *medellinus*, *gundlachi*, (accepting Simpson's interpretation of this latter species), and others.

PSORONAIAS, Crosse and Fischer (1893). This group of remarkable shells, embracing *crocodilarum*, *psoricus*, *semigranosus*, etc., was provisionally treated by Simpson as a group of *Elliptio*, but their remarkable sculpturing, and the deep beak cavities of some of their species, led him to observe that it was possible that the group should, after all, be placed in *Quadrula*.

I follow von Martens, in giving generic rank as above to the group. The type is *Unio psoricus*. To this genus we are enabled to add a species hitherto undescribed, under the name of

PSORONAIAS KUXENSIS, n. sp., Pl. VII, figs. 1, 2.

Shell small, compressed, rough, brown, biangular. Length 50, height 30, diam. 17 mm.

Shell hyperbolically rounded before, the extreme frontal point below the centre. Dorsum slightly arched, descending behind the ligament to the widely biangular posterior; the upper angle of which is midway the height, the lower angle very little above the base, which is nearly straight. The beaks are small, low, acute, approximate; and *apparently*, concentrically ridged. Epidermis dark brown (olivaceous and obsoletely rayed in the young), rough, the lines of growth numerous and well impressed. The discs are covered with fine pustulations, more pustular in front, biradially linear behind. The post ridge is low, but distinctly double, making the shell biangulate behind. The teeth are double in the left valve, single in the right. The cardinals are deeply sulcate and stout. Laterals slightly curved or nearly straight, separated by an interdentum. Nacre purple, beak cavities rather deep. Dorsal scars numerous, extending in a row from above the centre of the cavity down and forward upon the base of the cardinal teeth. Three well impressed muscle scars in front, two behind, the latter almost confluent. Habitat, Kux Creek, Chama, Guatemala. Collected by Mr. A. A. Hinkley, Feb. 6, 1917. A few dead specimens were obtained on the bank of the Isaibha River (Chama) of which the Kux Creek is a tributary. Type in Academy Natural Sciences. Cotypes in collection of A. A. Hinkley, the author and U. S. N. Museum.

I place this species in *Psoronaias* Crosse and Fischer, type *U. psoricus*, because of its evident relationship to *crocodilarum*, and *distinctus*, differing mainly from the latter in size and degree of inflation, being much inferior in both respects to *distinctus*.

ON THE RATE OF GROWTH OF POND UNIOS.

BY L. S. FRIERSON.

During the latter part of March 1916, the writer, for the purpose of constructing a fish pond, excavated a barrow-pit near the bank of a small creek, about ten feet wide, and at the time nearly dry. The barrow-pit was perhaps one hundred feet long, fifty feet wide and three feet deep. Early in April, 1916, the pit became full of water, overflowing from the adjacent creek, and together with two subsequent overflows, supplemented with seepage from the newly constructed fish pond, the pit remained more or less full of water, until May 25, 1917, when it was drained by a ditch into the nearby creek. From the dried bottom of this pit some thirty Unios were picked up by the writer. Ten of these were *Unio tetralasmus* Say, and the rest were *T. texasensis* Lea. All the specimens were of remarkably uniform size and appearance. The *texasensis* being about one and a half inches, and the *tetralasmus* two and a half inches long. Exact dimensions of a *texasensis*: length 43, height 24, diam. 16 mm.; of *tetralasmus* 75, 40, and 25 mm.

Both of these species had attained puberty. A female *texasensis* had its gills fairly full of young glochidia. A *tetralasmus* had several (three or four) ovisacs with a few (remaining) glochidia. In assigning an age to these shells it is quite sure that the *tetralasmus* discharges its glochidia in March and early April, so that when picked up on May 25, these shells were just about fourteen months old, from the date of discharge from their mother's gills.

In the case of the *texasensis* (which spawns somewhat later) it is possible that these were dropped by fish of which, at least six species obtained access to the pit on May 7, 1916 (on which date an overflow occurred), thus making about thirteen months. At any rate the maximum age of either

species is fourteen months from their mother's ovisacs. One of the *U. tetralasmus* is shown of natural size in Pl. VII, fig. 4.

Another observation concerning pond mussels might here prove of interest. A large pond was cut into two by a railroad embankment, a culvert preserving the level and providing communication between the two. In the lower and larger pond a half-bushel of Yonkapin (*Nelumbium luteum*) seed was sown. It was six years before these seeds germinated. These plants, during the summer, cover the entire surface of the pond with their broad peltate leaves. In this pond the writer planted a colony of a dozen *Anodonta grandis*. Several years after, taking advantage of extreme low water, the writer made a careful survey of these twin ponds, with the result that hundreds of Anodons could be found in the upper pond, but not a single one was found in the lower pond. Either the shade killed the young shells, or else the glochidia-laden fish avoided the shade of the lotus plants and congregated in the upper pond (there are no *Nelumbii* in the upper pond). Is not this avoidance of shade a reason for the paucity of unios in the tropics?

A NEW SOUTH AFRICAN NESOPUPA.

BY H. A. PILSBRY.

NESOPUPA FARQUHARI, n. sp.

Among Pupillidae sent by Mr. J. Farquhar there is a new species from Grahamstown which may be defined by comparison with *Nesopupa griqualandica* (Melv. and Pons.).^[5] The new form is ovate, of about the size of the other species, which it resembles in sculpture and in the lamellae of the parietal wall and columella. The two palatal plicae are subequal, the upper emerging to the lip, the lower one also long, reaching to the inner edge of the peristome. There is a very small nodule on the base of the columella. In *griqualandica* the lower palatal plica is short and very deeply immersed and there is a distinct though small basal plica within the base, in front of the lower palatal plica. In *griqualandica* there is a deep sulcus outside, over the upper palatal plica, and a flattening or short groove over the lower palatal; but in *farquhari* the sulcus is far less impressed except quite close to the lip. The color is reddish brown. Length 1.65, diam. 0.9 mm.

Mr. Burnup's figure 9, in Melvill and Ponsonby's Revision,^[6] may perhaps represent this species, while their description in the same paper appears to comprise both *griqualandica* and *farquhari*, though chiefly relating to the former. Their pl. I, figs. 8 and 10 represent *griqualandica*. The new form is named in honor of one of the most successful South-African collectors. It will be figured in the Manual of Conchology.

A NEW GUNDLACHIA FROM GUATEMALA.

BY BRYANT WALKER.

GUNDLACHIA HINKLEYI, n. sp., Pl. I, figs. 10–16; Pl. III, fig. 1.

Shell subovate, being much wider posteriorly, the anterior margin rather shortly rounded, the right margin nearly rectilinear, but somewhat diverging anteriorly, the left margin obliquely expanded and broadly rounded, anterior margin wider and much more curved than the posterior; apex very excentric, depressed and decidedly turned toward the right side, bluntly rounded, smooth except for a few concentric wrinkles; color a very pale corneous, nearly pure white; lines of growth rather strong and irregular; anterior slope with strong radial striæ originating below the septate growth and extending to the anterior margin, similar striæ appear on the left lateral slope, but are scarcely, if at all, visible on the right slope; the septate portion of the shell is small in comparison with the adult expansion, it is narrow and the posterior portion covered by the septum is free from and projects over, but scarcely beyond, the posterior margin of the adult aperture; the first growth of the shell from the septate form is continued on the sides in a nearly direct continuation of the lateral slopes of the septate shell for some little distance, the anterior slope of this stage is also a continuation of the anterior slope of the septate stage but owing to the oblique position assumed by the septate shell is at first somewhat convex, as viewed laterally, later as the side slopes begin to expand, the anterior slope is continued in a nearly straight line to the margin; the left lateral slope of the adult shell below the secondary constriction is concave at first, becoming nearly straight toward the margin; the right lateral slope is less concave above and straighter and more oblique than the left; owing to the small size of the septum and consequent large aperture of the septate shell and the narrow first growth of the adult shell there is no distinct aperture to the septate portion visible in the adult shell from below, the whole interior of the adult shell appears to pass, practically unconstricted, directly into the septate portion; the posterior margin of the adult shell narrow and somewhat abruptly expanded and reflected.

Length 5.5; width 3.75; alt. 1.75 mm.

The septate shell is oblong, the sides being nearly parallel, but slightly expanding anteriorly, the right slightly convex and the left slightly concave; the posterior margin is regularly rounded; the anterior more broadly

rounded; the apex depressed, bluntly rounded, excentric, reaching nearly to the right margin, smooth except for slight concentric wrinkles, lines of growth fine and regular; the anterior slope is slightly convex; the very short posterior slope below the projecting apex to the line of the septum is straight and oblique; the right lateral slope is steep and nearly straight, the left slope very convex; the septum is very short, being less than half of the length of the septate shell, convex on its lower surface, the margin is very short, being less than half of the length of the septate shell, convex on its lower surface, the margin is very concave and on the right side, extends further forward than it does on left, there does not seem to be the distinct thickening of the margin so noticeable in other species; aperture much larger than in any other species yet described.

Length 2; width 1.5; alt. .75 mm.

Type (43455 Coll. Walker) from the Maya Farm, Quirigua, Guatemala, collected by A. A. Hinkley. Cotypes in the collection of Mr. Hinkley.

This fine species is the first from either Central or South America, of which both the septate and adult forms are known.

It differs from all other described species except *crepidulina* Guppy in the small size of the septum and the consequent difference in the position of the aperture of the septate stage in the adult shell. The septum in the specimen figured appears very like the incomplete septum in the North American species, but as the three adult specimens before me are exactly alike in the position of the septate shell, this would seem to be the normal condition in this species. The specimen figured, which is 3.25 mm. in length, has apparently slightly passed the septate stage and begun the growth of the constricted portion of the adult shell and shows the beginnings of the radial striæ.

With the *Gundlachias* was associated a species of *Lævapex*, very like the *excentricus* Morelet. Whether it has any closer relations with the *Gundlachia* remains to be determined as the radula has not yet been examined. While the general aspect of the two species, if such they be, is very similar, the *Lævapex* has a very much more acute apex than the *Gundlachia*.

As shown by the figure, the radula of this species is quite typical of the genus.

A LIST OF SHELLS FROM THE EAST COAST OF FLORIDA.

BY BRYANT WALKER.

The late Dr. Charles A. Davis, the well known peat-expert of the U. S. Bureau of Mines, in addition to his special acquirements in geology and botany, was a good all-round zoologist and had a lively and unaffected interest in the work that any of his friends might be carrying on in that department. It was his kindly habit in his travels about the country to preserve any specimens that he came across that seemed to him likely to be of interest to any of his zoological friends. It will be remembered that the conchologists owe to him the rediscovery of the long lost *Planorbis multivolvis* Case, (NAUT., XXI, p. 16), and also the little *Lymnæa davis* Walker, (NAUT. XXII, p. 17), which bears his name.

In the spring of 1911 Dr. Davis' professional duties took him to Florida and while there he collected quite a number of samples of "drift," which in due time came into my possession. Several of the localities represented in the collection, such as Miami and St. Augustine, have already been reported upon by previous collectors and there seems to be no occasion to duplicate their work, but quite a number of the places visited by Dr. Davis have not been covered by any of the previous collectors in Florida and a record of the species found by him seems worthy of publication as a contribution to the distribution of the Mollusca along the east coast of the state.

I am indebted to Dr. George H. Clapp for the identification of the *Gastrocoptas* and *Vertigos*.

The list of localities and species represented in the collection is as follows:

MARSHES NEAR CHESTER SHOALS.

Euglandina rosea Fér.
Polygyra cereolus carpenteriana Bld.
Zonitoides minuscula (Binn.).
Zonitoides minuscula alachuana (Dall).
Vitrea dalliana ('Simpson' Pils.).
Pupoides modicus (Pfr.).
Gastrocopta rupicola (Say).
Gastrocopta pellucida hordeacella (Pils.).
Gastrocopta tappaniana (C. B. Ads.)?
Gastrocopta pentodon (Say).
Vertigo milium (Gld.).
Melampus coffeus (L.).
Detracia bulloides (Mont.).
Chondropoma dentatum (Say).
Plecotrema cubense (Pfr.).
Blauneria heteroclita (Mont.).
Microtralia minuscula (Dall).
Truncatella clathrus Lowe.
Truncatella caribæensis pulchella Pfr.
Truncatella bilabiata Pfr.
Littoridina monroensis (Ffld.).

Paludestrina? sp.? A single specimen that I can not approximate to any of the described species.

This is the first record, I believe, for *Plecotrema cubense* from the mainland of Florida. Originally described from Cuba, it was listed from the Bermudas by Dr. Pilsbry in 1900, (Trans. Conn. Acad., X, p. 504, pl. lxii, fig. 11), and there figured by him for the first time. Both he and Mr. John B. Henderson inform me that they have collected it on several of the Keys and I am indebted to both of them for the opportunity of comparing my specimen with theirs.

CHESTER SHOALS REFUGE STATION.

Euglandina rosea (Fer.).
Polygyra auriculata (Pfr.).
Polygyra uvulifera (Shutt.).
Polygyra cereolus (Mühlf.).
Polygyra cereolus carpenteriana (Bld.).
Polygyra cereolus septemvolva Say.
Polygyra cereolus volvox (Pfr.).
Praticolella jejuna (Say.).
Melampus coffeus (L.).
Detracia bulloides (Mont.).
Lymnaea humilis Say.
Physa cubensis Pfr.
Planorbis tumidus Pfr.
Planorbis alabamensis Pils.
Chondropoma dentatum (Say).
Truncatella bilabiata Pfr.
Truncatella clathrus Lowe.
Littoridina monroensis (Ffld.).

BETWEEN CHESTER SHOALS AND CAPE CANAVERAL.

Polygyra cereolus (Muhlif.).
Polygyra cereolus carpenteriana (Bld.).
Polygyra cereolus volvox (Pfr.).
Polygyra uvulifera (Shutt.).
Zonitoides minuscula (Binn.).
Pupoides modicus (Pfr.).
Gastrocopta pentodon (Say).
Gastrocopta pellucida hordeacella (Pils.).
Gastrocopta rupicola (Say).
Melampus coffeus (L.).
Detracia bulloides (Mont.).
Blauneria heteroclita (Mont.).
Chondropoma dentatum (Say).
Truncatella bilabiata Pfr.

Truncatella clathrus Lowe.

Truncatella caribæensis pulchella Pfr.

Amnicola. sp.? A single immature specimen.

CANAVERAL P. O.

Euglandina rosea (Fer.).

Polygyra cereolus septemvolva Say.

Polygyra cereolus volvox (Pfr.).

Polygyra cereolus carpenteriana (Bld.).

Pupoides modicus (Pfr.).

Helicina orbiculata Say.

EAU GALLIE.

Polygyra cereolus septemvolva Say.

Polygyra cereolus volvox (Pfr.).

Polygyra uvulifera (Shutt.).

Physa cubensis Pfr.

Helicina orbiculata Say var.

ISLAND OF EAU GALLIE.

Polygyra uvulifera (Shutt.).

Praticolella jejuna (Say).

Lymnaea humilis Say.

Physa cubensis Pfr.

PALM BEACH.

Euglandina rosea (Fer.).

Polygyra auriculata Say.

Polygyra cereolus carpenteriana (Bid.).

Strobilops floridana Pils.

Strobilops hubbari (Brown).
Pupoides modicus (Pfr.).
Gastrocopta contracta (Say).
Gastrocopta rupicola Say.
Gastrocopta pellucida hordeacella (Pils.).
Gastrocopta pentodon (Say).
Vertigo milium (Gld.).
Vitrea dalliana ('Simp.' Pils.).
Vitrea indentata (Say).
Zonitoides arborea (Say).
Zonitoides minuscula (Binn.).
Zonitoides minuscula alachuana (Dall).
Guppya gundlachi (Pfr.).
Thysanophora granum (Streb.).
Physa cubensis Pfr.
Helicina orbiculata Say.

LONG KEY.

Euglandina rosea (Fer.).
Polygyra cereolus (Mühlf.).
Polygyra cereolus carpenteriana (Bid.).
Strophia incanum (Binn.).
Pupoides modicus (Pfr.).
Gastrocopta pentodon (Say).
Gastrocopta rupicola (Say)?
Gastrocopta pellucida hordeacella (Pils.).
Gastrocopta pellucida hordeacella (Pils.) var. Small form.
Thysanophora incrustata (Gld.).
Thysanophora granum (Streb.).
Thysanophora dioscoricola (Guppy).
Guppya gundlachi (Pfr.).
Varicella gracillima floridana Pils.
Succinea campestris Say?
Melampus coffeus (L.).
Detracia bulloides (Mont.).
Microtralia minuscula (Ball).

COLLECTING IN DIGBY, NOVA SCOTIA.

BY LILLIAN DYER THOMPSON.

While traveling through Nova Scotia and New Brunswick last summer, we stayed for about six weeks at Digby, N. S. Digby is about 200 miles northeast of Boston, and is situated near the Bay of Fundy, opposite St. John, N. B. The town is located on the southeast shore of the Annapolis Basin,—a sheet of water about twenty miles long and ten miles wide. This basin is connected with the Bay of Fundy by a channel about three-fourths of a mile wide at its greatest width. This channel, known as Digby Gap, is noted for its rapid tides,—the rate of flow through the Gap being about eight miles an hour. The tide fall at Digby is thirty feet. The shores of the Basin are sandy, with the exception of the two rocky promontories on each side of the Gap; the one which is nearest to Digby being Point Prim. The town is on a small peninsula on either side of which are two inlets of the Annapolis Basin, known as the Racquet, on the west, and the Jacquet, on the east of Digby proper. On the ebb tide these are almost dry, exposing long mud flats.

There is one island in the Basin, about opposite the Gap and at the mouth of Bear River, called Bear Island. From this a long bar extends, called Bear Island Bar, which is covered to a depth of about six feet at low water, and is covered with eelgrass.

Near the Yacht Club pier were found many *Polinices heros*, and their red-brown “sand-collars.” In the Jacquet were many *Litorina littorea* and *Litorina rudis*. On the exposed beach, nearer the town, we found *Mytilus edulis*. On the rocks, in the Racquet, we found *Thais lapillus* and a host of *Acmaea testudinalis* ranging in size from one-eighth of an inch to about an inch in diameter. In the mud, at the base of the rocks, were a multitude of *Buccinum undatum*, *Neptunea decemcostata*, ranging in size from one-

eighth of an inch to about an inch in diameter. In the mud, at the base of the rocks, were a multitude of *Buccinum undatum*, *Neptunea decemcostata*, and *Colus stimpsoni*, all alive and half-buried. Some dead specimens of *Aporrhais occidentalis* were also found, five of them being full-grown.

On the suggestion of Capt. Danforth, we constructed a dredge, and endeavored to dredge Bear Island Bar from his motor-boat. Here we found quantities of *Lacuna vincta*, *Alectrion obsoleta*, *Cylichna alba*, and two *Polinices triseriata*.

There were some soldiers encamped at Digby, and they used to gather *Litorina littorea* and steam and eat them, without any flavoring. They sometimes ate *Thais lapillus* also. One day, after a rain, we found two *Helix hortensis* crawling along the road.

A NEW TYPE OF THE NAYAD-GENUS FUSCONAIA. GROUP OF F. BARNESIANA LEA.

BY A. E. ORTMANN.

During the study of the nayad-fauna of the upper Tennessee, the present writer found that there exists, in this region, a peculiar type of shells, belonging to the genus *Fusconaia*, the various forms of which have been described previously under a great number of specific names, which, however, seem to belong all to one species. In addition, among material received from L. S. Frierson from the Ozark Mountains, a form was discovered which presented the same structure.

The oldest name for the upper Tennessee form is *Unio barnesianus* Lea. A more detailed account of its various phases is to be given elsewhere, and it suffices here to mention only those forms which belong here. According to obesity, I distinguish three local, or ecological races:

1. FUSCONAIA BARNESIANA (Lea) 1838.

U. barnesianus Lea, '38. *U. meredithi* Lea, '58. *U. pudicus* Lea, '60. *U. Lyoni* Lea, '65. *U. tellicoensis* Lea, '72. *U. lenticularis* Lea, '72.

As the normal (most abundant) forms we may regard *U. meredithi*, *pudicus* and *lenticularis*, which differ from each other only in the development of the rays (topotypes examined). *U. barnesianus* is a slightly more elongated individual, with poorly developed rays. *U. tellicoensis* (topotypes examined) is a *lenticularis* slightly more swollen; and *U. Lyoni* forms the transition toward var. *tumescens*, having a little more elevated beaks, greater obesity, and rather distinct rays.

2. FUSCONAIA BARNESIANA BIGBYENSIS (Lea) 1841.

U. bigbyensis Lea, '41. *U. estabrookianus* Lea, '45. *U. fassinans* Lea, '68. *Pleurobema fassinans rhomboidea* Simpson, '00.

The most frequent form is *fassinans rhomboidea* (topotypes examined), with rays poorly developed. *U. bigbyensis* has more distinct rays; *U. estabrookianus* (topotypes examined) is an old, overgrown form, without rays; *U. fassinans* is founded upon an individual (type examined, also topotypes), which is exceptionally elongated, without rays.

3. FUSCONAIA BARNESIANA TUMESCENS (Lea) 1845.

U. tumescens Lea, '45. *U. crudus* Lea, '71. *U. radiosus* Lea, '71.

U. tumescens is the most typical form, greatly swollen, with more or less developed rays; *U. radiosus* (type and topotypes examined) is less swollen, but for the rest like *tumescens*; *U. crudus* (topotypes examined) lacks rays, and has much eroded beaks, but stands close to *radiosus*.

The mutual relations of these forms may be understood by the help of the following key. Only the three largest divisions are to be regarded as varieties, in the other forms the characters are merely individual, although specimens representing only one (or a few) of these "forms" often prevail at a given locality.

a₁. Flat, compressed, dia. of shell less than 40 per cent of the length (var. *bigbyensis*).

b₁. No rays, or rays obscure, color of epidermis brown, dark.

c₁. Rhomboid in shape.

d₁. Large. *Estabrookianus*.

d₂. Smaller. *Fassinans rhomboidea*.

c₂. More ovate, tapering behind. *Fassinans*.

b₂. Rays distinct, well developed over most of the disk. Ground color of epidermis lighter. *Bigbyensis*.

a₂. Moderately convex, dia. 40–49 per cent of length. *Barnesiana typica*.

b₁. Beaks not elevated, shape trapezoidal, rhomboid, or subovate.

c₁. Dia. about 41 or 42 per cent; size small.

d₁. Shape somewhat elongate (trapezoidal); rays obscure.
Barnesianus.

d₂. Shape shorter (rhomboidal).

e₁. Rays obscure. *Lenticularis*.

e₂. Rays present, color of epidermis lighter.

f₁. Rays few. *Meredithi*.

f₂. Rays numerous. *Pudicus*.

c₂. Dia. about 45 per cent; larger. Shape subovate. Rays obscure.
Tellicoensis.

Lyoni.

b₂. Beaks more elevated, shape subtriangular. Dia. 46 per cent, with rather distinct rays. *Lyoni*.

a₃. Much swollen, dia. over 50 per cent. Beaks elevated. (var. *Tumescens*).

b₁. Without rays. Dia. 51 per cent. Beaks much eroded. *Crudus*.

b₂. With rays. Dia. about 56 per cent or more.

c₁. Dia. about 56 per cent. *Radiosus*.

c₂. Dia. about 64 per cent. *Tumescens*.

As to the geographical distribution, it should be briefly stated that the swollen forms (a₃) inhabit the largest rivers; the compressed forms (a₁) are found in the headwaters, and the intermediate forms (a₂) belong to the streams of moderate size. Intergrades are frequent.

ANATOMY.^[7]

All these shells have the same, and an extremely characteristic and unique structure of the soft parts, so that there is not the slightest question that they belong together. I have examined the soft parts of some 200 specimens in the field, and over three dozens have been preserved in alcohol, and have been examined at leisure in the laboratory. They include representatives of the three main varieties, and of practically all of the individual variations.

Gravid females have been found on the following dates: May 11, '13; May 15, '13; May 16, '13; May 20, '13; May 20, '14; May 22, '14; May 25, '14; July 5, '13; July 9, '13; July 10, '13; July 13, '13; July 14, '13. *Glochidia* have been observed on May 20, '14 (immature), and July 14, '13. Thus this species evidently is a summer breeder (tachytictic).

The soft parts are those of the genus *Fusconaia*: the *supraanal* is separated from the *anal* by a very short mantle-connection, which is absent (or torn?) in rare cases. Inner lamina of inner gills free from abdominal sac. All four gills are marsupial. Placentae well developed and subcylindrical.

Branchial opening with well developed papillae, *anal* with distinct, but small papillae. *Palpi* subfalciform, posterior margins connected at base only.

While thus the *Fusconaia* structure is typically developed, this species is quite unique in its color. This concerns chiefly the color of the gonads, eggs, and placentae.

The soft parts are often uniformly pale, whitish, but may shade to orange, and the orange is most prominent on foot, adductors, and mantle-margin; but the paler tints prevail, and often the orange is replaced by yellowish or brown. The gills are pale, but are generally suffused with blackish. The gonads are brown to red, mostly of a peculiar dull lavender color in the female, and the latter color, or purplish brown, is the prevailing color of the eggs and placentae. The charged gills become thus rather dark purple, or purple-brown, shading sometimes to dull red or blackish, in other cases to brownish, brownish pink, brick-red, or even pale brown. These are very peculiar tints, by which this species is easily recognized in the field: four

marsupial gills of this blackish-purple color are not known in any other *Nayad*.

Glochidia have been found only in specimens belonging to the headwaters variety (*barnesiana bigbyensis*). They are subelliptical, slightly higher than long, L. 0.15, H. 0.16 mm.

Although a true *Fusconaia*, this species (or group of forms) stands isolated within the genus, in characters of the shell as well as in the soft parts. It differs from the species of the *subrotunda*-group (ind. *ebena*, *pilaris* etc.) very markedly by its smaller size and by the very shallow beak cavities. The forms of the *undulata*-group (incl. *flava*, and the *cuneolus*- and *corforms*) have generally also somewhat deeper beak cavities, and the shell has a more or less distinct posterior ridge, with a flattening or a shallow groove in front of it, characters which are missing in the *barnesiana*-group. As has been pointed out, in the latter group, the color of eggs and placentae is remarkable: in all other forms of *Fusconaia*, this varies from white to bright red.

I introduce here another species, in order to show that the *barnesiana*-type is also represented outside of the Cumberland-Tennessee drainage, namely in the Ozarks.

FUSCONAIA OZARKENSIS (Call) 1887.

F. ozarkensis Call, Pr. U. S. Mus. 10, '87, p. 499, pl. 27. Tr. St. Louis Ac. 7, '95, p. 33, pl. 18. *Lampsilis ozarkensis* Meek & Clark, Bur. Fisher. Doc. no. 759, '12, p. 18. *Pleurobema utterbacki* Frierson, in: Utterback, Naiad. Missouri (Amer. Midland Natural 4, 1916, p. 86, pl. 5, pl. 20, f. 63).

I have specimens from James River, Galena Stone Co., Mo., and White River, Cotter and Norfolk, Baxter Co., Ark., donated by L. S. Frierson and collected by A. A. Hinkley on July 30 and Aug. 2 and 5, '14, A number of specimens (8) were preserved in alcohol, coll. July 30 and Aug 2, which all were gravid females, and one of each date had glochidia. This marks probably the end of the breeding season, and the species is tachytictic.

There is some confusion with regard to this species. After the first description by Call, it has not again been recorded, except by Meek and Clark, and I believe, the identification of these authors (supported by B. Walker) is correct. But I think that other authors have seen this form, but have not recognized it, and, for instance, Simpson's *pannosus* and *subellipticus* (regarded as varieties of *Pleurobema argenteum* and *breve* respectively) are also this. Frierson's *utterbacki* is surely this, since my specimens were thus labeled by Frierson.

Walker, Frierson, and Simpson (in part) believe this to be a *Pleurobema*, and not a *Lampsilis* (see also Simpson, '00, p. 557, and '14, p. 131), and this comes nearest to the truth, in fact, it is the most plausible assumption to be made from the study of the shell alone. The shell "resembles a very elongated *Quadrula coccinea*," according to Meek and Clark, and the comparison with *Pleurobema argenteum* and *breve* (which, by the way, are synonyms), made by Simpson, is significant. We must keep in mind that Call's fig. 4 represents the normal shape of the shell, while his fig. 1 is rather abnormal, and possibly does not belong here at all. These two figures by no means represent the female and male, as Call believes.

The investigation of the soft parts has shown that this actually is a *Fusconaia*. Corresponding, both in soft parts and shell, to the *barnesiana*-type of the upper Tennessee region. *F. ozarkensis* differs from *barnesiana* by the more elongated (subtrapezoidal) outline of the shell, more anterior

beaks, and the weak development of the rays, which are faint at the best, and often entirely absent. A swollen form of it is not known to me, but specimens from White River are slightly more convex than those from James River (farther up). Also Utterback's quotation of Frierson (p. 87, footnote) make it probable that there are differences in obesity.

ANATOMY.

Supraanal opening probably separated from the *anal* by a short mantle-connection, but in all my specimens this is torn by rough handling. Inner lamina of inner gills free from abdominal sac. All four gills marsupial in the female. *Placentae* well developed and *subcylindrical*.

Anal opening with small papillae, branchial opening with well developed papillae. *Palpi* as usual, their posterior margins connected for about one third of their length or less.

As to the color of the soft parts, which is so characteristic in *barnesiana*, not much can be said, since my material has been too long in alcohol. But in most of my specimens the gills are yet distinctly suffused with black. The *placentae* have been rendered whitish, but here and there traces of a dark stain are preserved (which is disappearing gradually). It is quite possible that the color of the *placentae* originally was similar to that of *barnesiana*.

The *glochidia* are subelliptical, slightly higher than long; L. O. 15, B. O. 18, thus agreeing with those of *F. barnesiana*.

NOTE ON THE RELATION OF SNAIL FAUNA TO FLOODS.

BY A. RICHARDS.

During the years 1911 to 1916, while the writer was a member of the faculty of the University of Texas, a series of incidental observations on the snail fauna of Waller Creek was made. These observations have now come to an end due to the change of residence of the observer. It seems not unwise, therefore, to publish a short note on the subject in the hope that the facts recorded, although fragmentary, may have a bearing on the work of some other follower of snail life.

Waller Creek is a small stream near the University of Texas at Austin. It is some four miles in length and empties into the Colorado River at a distance of perhaps two miles below the University. That portion of the stream close to the University between Fifteenth and Twenty-seventh Streets, was most closely observed, but data was also collected from the region below. During the hot months, from about July 1st to October 1st usually, the stream is dry, or water is to be found only in an occasional pool; during the rest of the year the water flows to a depth of a few inches. The bed is scoured out of limestone (Austin Chalk) and has for much of its length a solid flat rock bottom. The banks of the creek have in general a gradual slope. In time of flood and during heavy rains, this stream rises very rapidly and quickly becomes bank-full, so that the water rushes down in a torrent, the roar of which may at times be heard for a distance of some blocks. The fall of the creek is considerable, being about 75 feet in two miles from Twenty-seventh Street to the Colorado, and this fall in connection with the shape of the bed gives to the current in times of flood a tremendous force.

During the first two years of this observation, 1912–1913, the snail population of the creek in its middle stretches was dense. There were in particular two species very thickly represented, *Planorbis lentus* and *Physa halei*. So numerous were they that one could in a few moments within a very few feet gather a pint of either kind. Wherever a little ripple or a tiny waterfall occurred were many snails oriented in relation to the current, their heads pointing into it. Elsewhere in the more quiet water they were also to be found, but in less numbers. These conditions obtained especially in the early spring; as the breeding season, which in that latitude extends over half the year, passes by, the snails of course become much less numerous.

It is to be noted that previous to the time when the snails had become so abundant, there had been no heavy rains of sufficient importance to be recorded since 1908. Excessive rains occurred in May 1908, November and May 1907, June and March 1905, May and April 1904, July and February 1903, July and November 1902. There was a very severe flood in the creek in April 1900. It will be seen that between the time when my observations began and the last excessive rain considerable time had elapsed and the snails had had the opportunity to reinstate themselves in the creek, assuming that they had suffered in those floods as they have done in the later ones.

In the fall of 1913 there were two floods of unusual proportions in the creek. In October it rose very rapidly, but shortly subsided, and in November, at the time when the entire state was visited by the most severe flood since 1869, it was again subjected to a very thorough scouring. Excessive rains fell on several consecutive days, and streams in the entire Colorado watershed were out of their banks.

After the heavy rains of the earlier part of the month there were several days upon which the rainfall, while comparatively light, was sufficient to keep the creek much higher than its normal level. When the water finally subsided to its normal amount the bed was covered completely with a layer of detritus and soft green humus and algae from a half an inch to an inch in thickness. This deposit and the acids formed from it have been the cause of a much more rapid disintegration of the limestone than had been the case in the immediately preceding years. Loose pieces of limestone which were exposed to the action of the water had in many cases fallen apart by the end

of January. Further rises occurred on April 27 and on May 20th, 1914, but were not sufficient to remove all of the accumulated layer of detritus.

In January 1914, a search for snails where they had before been numerous failed to reveal a single specimen of *Planorbis* and less than half a dozen *Physa*. Even in the deeper pools they could scarcely be found. Later in the spring in the lowest part of the stream a number of very small *Physa*, as well as some clusters of eggs were found. The force of the current had been so great as to wash the snails down to the river, and it is possible that the condition of the water due to the decomposing humus may have affected those which were able to escape the flood danger. That some of the *Physa* were left after the flood may be attributed to their pointed shape which decreases the amount of force that the water was able to exert on them as compared with that on the flat *Planorbis* shells. Except as noted above in April and May 1914, the conditions in the creek remained as normal.

In January 1915, *Physa* has again made its appearance in the middle parts of the stream, although in small numbers only. Diligent search, however, failed to reveal a single *Planorbis*. The snails which were present were found only under fair-sized rocks where they were well protected; they did not occur out in the open stream as had been the case when the creek was more densely populated with snails.

In April 1915, on the 22d and 24th, Waller Creek was swept by three scouring floods which devastated the entire bottoms. The water rose higher during the night than it had at any time since 1886; bridges were washed away and much damage done. The bottom of the creek was again washed clean except under the larger stones and in deep holes in the bed. Snails were not observed in any numbers following this flood during the rest of that year. Early in January 1916, however, *Physa* had again appeared fairly abundantly where they had formerly been very thick, in the region just above the University; later in the spring they became quite numerous here. In January of this year the first *Planorbis* were found that had been seen in the creek since the autumn floods of 1913. Between Fifteenth and Sixteenth Streets in a pocket containing good-sized stones over which the water flows rather swiftly a number of specimens were taken, although none were found above or below this locality. It is noted that below this region the creek is frequently covered with oily scum and that it receives the refuse from the adjoining properties. Except after high water which would clean it out, the

creek in this region would hardly be expected to support much snail life. It must be supposed either that the *Planorbis* had made their way up to Sixteenth Street during the short time following the floods before the water became badly contaminated, or else that in this particular place a few specimens from the previous years had withstood the floods and reproduced themselves in sufficient numbers to be noticeable in January. Of these two suppositions the latter seems much the more rational.

After the flood of 1915 the water subsided very quickly so that a new layer of humus and algae was not deposited, but that the bed was again restored to its former condition of a clean smooth rock bottom. Upon the return to this condition the snail population increased very rapidly, and at the time when the last observation was made in the early summer of 1916, seemed in a fair way to return to the condition of 1912.

It seems to the writer that the slow return of the snails during the year 1914 was due not only to the repeated rains of the fall and winter of 1913–14, but also to the condition of the creek bed at this time. Although the heavy rains of 1915 were more severe, they were confined to one month, and the creek bed was left in a very much cleaner condition than during the preceding year.

The conditions of life which obtain now in Waller Creek are those of a new life region. This must of necessity be so in an intermittent stream to a certain extent, but owing to the flood conditions here they are doubly so.

The chief facts of interest in regard to the habits of snails as shown by these observations, are these: 1, The snails of both species are to be found commonly in uncontaminated water which is running at a fairly rapid rate, and the most common orientation is with the heads pointing up stream; 2, The snail population in any stream is subject to wide fluctuations depending upon flood conditions; 3, *Physa* because of its shape is less affected by floods than *Planorbis*; 4, The return of the snail population to a given stream is determined not only by the frequency and severity of the floods, but also by the condition of the bottom of the stream after the subsidence of the high water; 5, The snails thrive best where there is a constantly renewed supply of clean water which contains little decaying vegetation.

Wabash College, Crawfordsville, Ind.

WILLIAM BULLOCK CLARK.

Dr. William Bullock Clark, professor of geology in the Johns Hopkins University, eminent for his contributions to geology, died suddenly from apoplexy on July 27, at his summer home at North Haven, Maine.

Wm. Bullock Clark was born at Brattleboro, Vermont, December 15, 1860. Since 1894 he was professor of geology in Johns Hopkins University. In 1896 Professor Clark organized the Maryland Geological Society, and has been State Geologist since that time. The admirable volumes on paleontology of Maryland, issued under his direction, are widely used by conchologists interested in fossil mollusks. This series of reports will be his enduring memorial. Professor Clark's chief paleontological interest was in the *Echinoidea*, upon which he published several monographs.

PUBLICATIONS RECEIVED.

A MONOGRAPH OF WEST AMERICAN MELANELLID MOLLUSKS. By Paul Bartsch (Proc. U. S. Nat. Museum, Vol. 53, pp. 295–356, pls. 34–39, Aug. 1917). This completes the discussion of the West American mollusks of the super-family Pyramidelloideae, comprising the family Pyramidellidae, which has been previously treated, and the Melanellidae here considered. The former are readily distinguished by having the “nepionic whorls sinistral and tilted; the axis of the early whorls usually being at right angles to that of the succeeding turns, in the first of which the nuclear whorls are frequently quite strongly imbedded.” In the latter the early whorls are dextral and never tilted or immersed. A review of the work done in this group is followed by the descriptions of the species, including forty-nine new species and one new genus *Eulimostraca*. The illustrations are excellent.

NEW AND LITTLE KNOWN SPECIES OF SOUTH AMERICAN MUSSELS OF THE GENUS DIPLODON. By William B. Marshall (Proc. U. S. Nat. Museum, Vol. 53, pp. 381–388, Pls. 50–55, August, 1917). Two new species *Diplodon felipponei* and *D. fortis* are described and figured, together with six species described by Mr. C. T. Simpson in his Descriptive Catalogue of the Naiades and not previously figured.

NOTES ON THE SHELLS OF THE GENUS EPITONIUM AND ITS ALLIES OF THE PACIFIC COAST OF AMERICA. By William H. Dall (Proc. U. S. Nat. Museum, Vol. 53, pp. 471–488, August, 1917). An interesting account of the various groups and subgenera is followed by descriptions of forty-two new species.

The name *Pictoscala* is proposed for a section, type *Scalaria lineata* Say.

STUDIES ON AUSTRALIAN MOLLUSCA. PT. XIII. By C. Hedley (Proc. Linn. Soc. New South Wales, 1916, vol. 41, pt. 4, pp. 680–719, pls. 46–52, issued April 4, 1917). The author's notes under *Tridacna gigantea* Perry are of such general interest that we quote them in part. "Under the name of *Chama gigas* the father of Natural History seems to have embraced the whole of the modern genus *Tridacna*. For the name *gigas*, as restricted to a single species, the candidates are the shell subsequently named *squamosa* by Lamarck and a huge species whose valves in the Ulrica Museum, together weighed 498 pounds.

"After careful examination, Hanley decided that the furbelowed clam, such as Reeve has figured for *T. squamosa*, ought rightly to bear the name *gigas*. He based his verdict on the ground that the actual shell owned by Linné as representing *gigas*, is the Lamarckian *squamosa*, and that to this apply most of the literary references. Linnean contemporaries such as Born, Regenfuss and Chemnitz, while making casual references to the giant, all agree in figuring and describing *squamosa* as the Linnean *gigas*.

"Discriminating in 1819 between the species his predecessors had confused, Lamarck unlawfully used the name *gigas* for the largest form, while for the Linnean *gigas* he proposed *squamosa*. Attentive to the remarks of Hanley, Hidalgo in 1903, renamed the biggest species *T. lamarcki*. But in 1811, Perry had already used the name *Chama gigantea* for 'the largest shell at present known.' As the young of the giant has not yet been traced to the adult, it is still possible that *squamosa* is a juvenile deeper-water form of the large intertidal and abraded *gigantea*.

"The heaviest known are a pair weighing 550 lbs., which Cuvier and Lamarck relate were presented by the Venetian Republic to Francis I. These still exist, their edges bound with brass, as holy-water basins in the cathedral of St. Sulpice, in Paris.

"The photographs of Saville Kent show the giant clams in their natural position on the Great Barrier Reef, where they occur free and exposed at low tide, standing on their umbones, and showing their brightly colored mantle and so-called eyes as they gape."

There are many other interesting notes bearing on nomenclature, and the animals of Australian species. Six new species are described and twenty-nine species figured.—C. W. J.

AN ANNOTATED LIST OF SHELLS FROM NORTHERN MICHIGAN. By Mina L. Winslow (Occasional papers, Mus. Zool., Univ. Mich., No. 42, July 1, 1917) a list of sixty-five species from Schoolcraft, Alger and Chippewa counties, also a list from Isle Royale.

NOTES.

THE OLDROYD COLLECTION.—Mr. and Mrs. J. S. Oldroyd have given their collection of shells to the Leland Stanford Jr. University, and are now permanently employed in the Museum, Mrs. Oldroyd being the curator. The collection has been placed in the Department of Geology and Mining. The Stanford alumni purchased the collection and library of the late Henry Hemphill, which, with the Law collection and several others, forms an unusually fine working series. Mr. and Mrs. Oldroyd have spent about eight weeks at Friday Harbor, Puget Sound and British Columbia making large collections for the Stanford University, California Academy of Science and University of California.

NORTH CAROLINA LAND SHELLS.—The following species of land shells were picked from leaf-mold collected at Spruce Pine, Mitchell Co., North Carolina, by Samuel G. Gordon while on a mineralogical excursion. The specimens are in the collection of the Acad. Nat. Sci., Phila., *Gastrocopta contracta* Say, *G. pentodon* Say, *Circinaria concava* Say, *Polita indentata* Say, *Taxeodonta lamellidens* Pils., *Gastrodonta elliotti* Redf., *G. guldaris* Say, *Euconulus sterkii* Dall., *Punctum pygmæum* Drap., *Carychium exile* Lea.—E. G. VANATTA.

A CORRECTION.—In my little paper, "Descriptions of New West American Marine Mollusks and Notes on Previously Described Forms," Proc. U. S. Nat. Mus., Vol. 52, pp. 670–671, plate 46, figure 2, 1917, I published *Cerithiopsis (Cerithiopsis) helena* from Panama, type Cat. No. 204128, U. S. N. M. Mr. Vignal, of Paris, has been kind enough to call my attention to the fact that the same combination was used by O. Boettger in 1901 for a fossil in his contribution "Zur Kenntniss der Fauna der mittelmioocänen

Schichten von Kosteĵ im Krassò-Szörényer Komitat,” in “Verhandlungen und Mitteilungen des siebenbürgischen Vereins für Naturwissenschaften zu Hermannstadt,” p. 128, 1901. It is therefore necessary to bestow a new designation on my shell, and it may be known as *Cerithiopsis (Cerithiopsis) anaitis*.—PAUL BARTSCH.

ALBINISTIC EPIPHRAGMOPHORA FIDELIS.—A few days ago, I had two hours in Gladstone Park. The Park is wooded, and there are large moss-covered rocks. *E. fidelis* was out freely. I found one light one. The one sent you some years ago was uniform in color. This one shows two distinct dark narrow bands around the lower whorl, but not showing anywhere else. This is the fourth one I have found in the Park in twenty years or more, though I have been there often.

—J. G. MALONE, *Portland, Ore.*

The many friends of DR. HERMANN VON IHERING will hear with deep regret that he has been removed by political intrigue from his position of Director of the MUSEU PAULISTA at Sao Paulo, Brazil. This museum was founded by Dr. von Ihering. His eminence as a zoologist and unceasing activity as an investigator of the South American fauna, had won for it an honorable place among scientific institutions. We understand that his successors are men without knowledge of the biological sciences. As the only scientific assistant, Mr. Rudolph von Ihering has resigned, it appears that the scientific activity of the State Museum of Sao Paulo has come to an end—a real calamity to American zoology and paleontology.

Dr. von Ihering is located at present at Hansa de Joinville, State of Santa Catharina. He is in good health, and is engaged in the preparation of his work: “Die biogeographischen Grundgesetze,” several chapters of which will deal with mollusks.—H. A. P. & C. W. J.

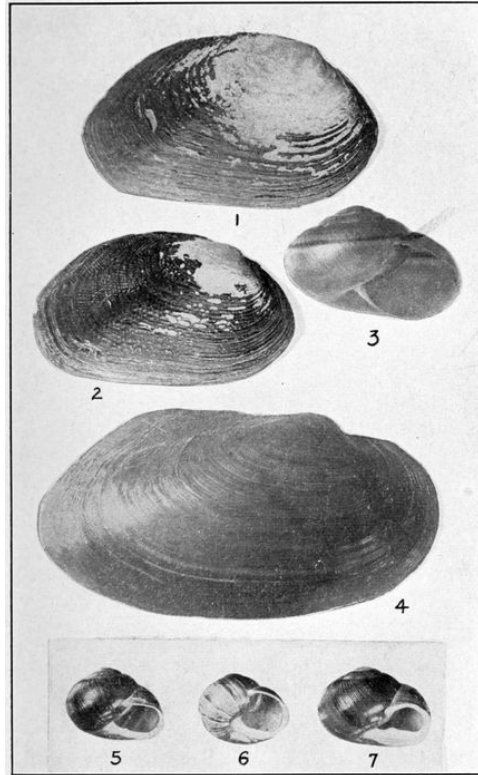


PLATE VII

- 1, 2. PSORONAIAS KUXENSIS FRIERSON.
3. EPIPHRAGMOPHORA CALLISTODERMA PILS. & FERR.
4. UNIO TETRALASMUS SAY.
- 5, 6. ZACHRYSIA RAMSDENI PILS.
7. Z. EMARGINATA PFR.

[1.](#) NAUTILUS, vol. xx, pp. 58–60.

[2.](#) NAUTILUS, xxvi, pp. 51–55, 1912.

[3.](#) In a later work Ortmann classifies both *ventricosa* and *cohongoronta* as varieties of *ovata* Say.

[4.](#) The relations of mollusks to fish in Oneida Lake. By Frank Collins Baker. Technical Publication No. 4, New York State College of Forestry at Syracuse University. Pp. 366. Syracuse, N. Y., 1916. We are informed that it may be obtained free by those interested in the study of Mollusca by applying to the dean of the college, Dr. Hugh P. Baker.

5. *Pupa griqualandica* M. and P., 1893; the specimens used being from Pretoria.

6. Ann. Mag. N. H. (8), i, p. 76, pl. i, 1908.

7. In NAUTILUS, 28, 1914, p. 31, I have described the anatomy of “*Pleurobema fassinans*.” This is a mistake: the shells examined belong to *Pleurobema* all right, but are the form known as *U. argenteus* Lea, which belongs to the *oviforme-group*, and should be called: *Pleurobema oviforme argenteum* (Lea). These will be treated more fully elsewhere.



TRANSCRIBER'S NOTES

1. Silently corrected typographical errors.
2. Retained anachronistic, non-standard, and uncertain spellings as printed.
3. Footnotes have been re-indexed using numbers and collected together at the end of the last chapter.

*** END OF THE PROJECT GUTENBERG EBOOK THE NAUTILUS.
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